Chapter 11. In conclusion: A review and roadmap

In this final chapter we'll first take a moment to briefly review what we've learned, highlighting and distilling what we think are the most important skills and concepts to take away. We have covered the fundamentals of *reinforcement learning* and if you have made it this far and have engaged with the projects, you're well-positioned to implement many other algorithms and techniques.

This book is a *course* on the fundamentals of *deep reinforcement learning*, not a textbook or reference. That means we could not possibly have introduced all there is to know about DRL, and we had to make tough choices about what to leave out. There are a number of exciting topics in DRL we wished we could have included, and there are some topics that, despite being "industry standards," were inappropriate to include in a project-focused introductory book like this one. However, we wanted to leave you with a roadmap for where to go from here with your new skills.

In the second part of this chapter, we'll introduce at a high-level some topics, techniques, and algorithms in DRL that are worth knowing if you're serious about continuing in the DRL field. We didn't cover these areas because most of them involve advanced mathematics that we do not expect readers of this book to be familiar with, and we did not have the space to teach more mathematics.

11.1. What did we learn?

Deep reinforcement learning is the combination of deep learning and reinforcement learning. Reinforcement learning is a framework for solving *control tasks*, which are problems in which an *agent* can take actions that lead to positive or negative rewards, given some *environment*. The environment is the universe in which the agent acts. The agent can either have full access to the state of the environment, or it may only have partial access to the state of the environment, which is called *partial observability*.